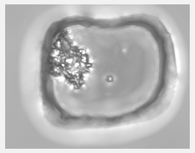
Phytolith Identification: from both Discriminative and Generative Perspectives

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Phytoliths

Phytoliths are microscopic, silica structures from plants. They provide a wealth of information to biologists, archaeologists, and anthropologists about past ecology





Ehrharteae, pictured above [1], is a tribe in the subfamily Oryzoideae. Both are part of the larger grass family, Poaceae. Below, the entire taxonomic hierarchy of the phytoliths in our dataset

Subfamily	Bambusoideae			Oryzoideae			
Tribe	Bambuseae	Arundinarieae	Olyreae	Oryzeae	Streptogyneae	Phyllorachideae	Ehrharteae
Genus	Otatea	...					

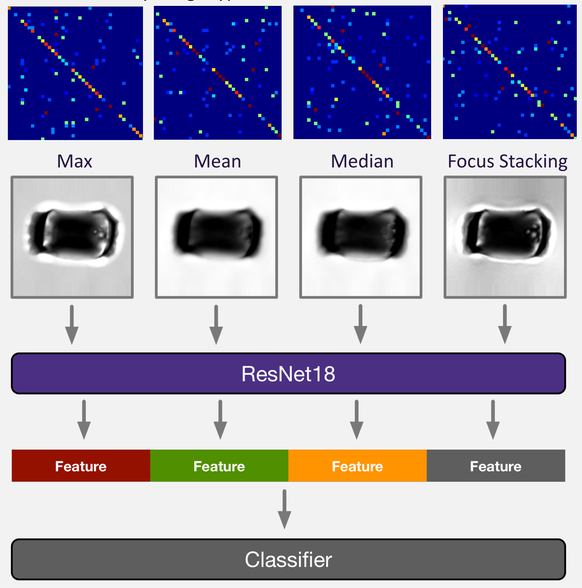
Currently biologists rely on expertise and complicated statistical methods to identify morphometries and classify phytoliths. We aim to:

- Develop a classifier that accurately and automatically learns from data to help classify genera
- Help biologists learn the morphological traits of different genera

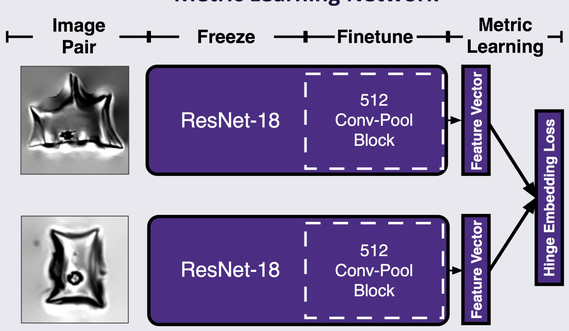
Discriminative Models

Multi Style Features

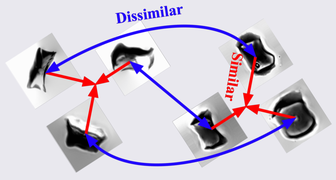
Confusion matrices highlight the different genera misclassifications that occurred by image type

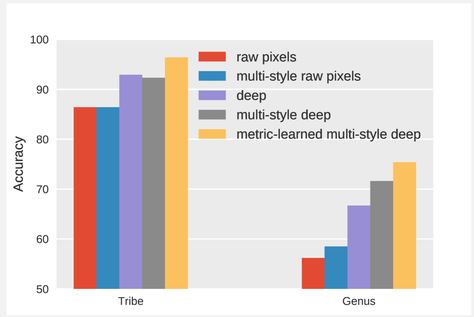


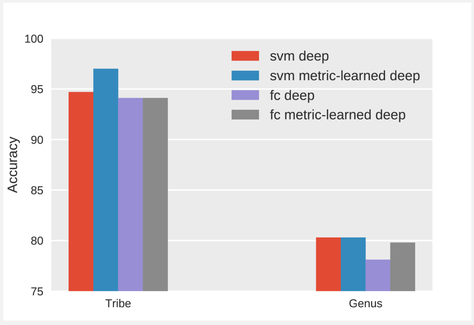
Metric Learning Network



$$L(\vec{F}_1, \vec{F}_2) = \begin{cases} \left\|\vec{F}_1 - \vec{F}_2\right\|_2 & \text{Similar} \\ \max\left\{0, M - \left\|\vec{F}_1 - \vec{F}_2\right\|_2\right\} & \text{Dissimilar} \end{cases}$$

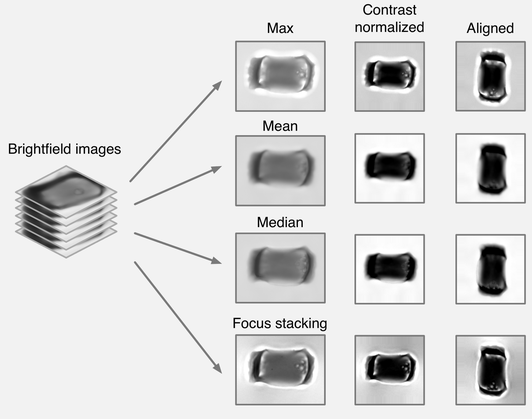




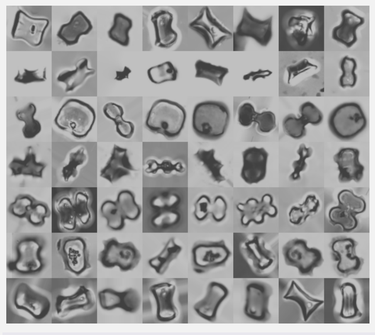


Data Processing

- Multiple merging strategies
- Autocontrast normalized
- Principal axes aligned using PCA

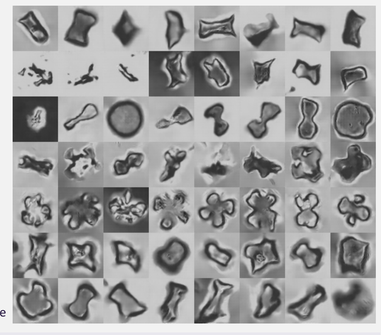


Generative Models



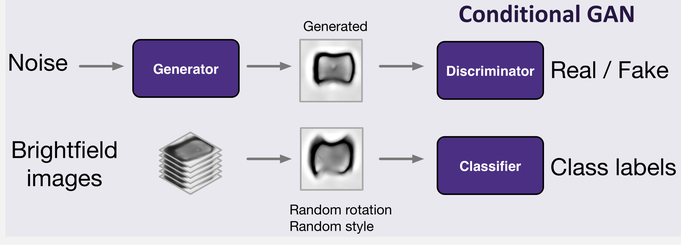
Real

Which is real?



Fake

Conditional GAN



To prevent mode collapse and instability:

- Random rotation
- Weighted loss
- Train using Wasserstein distance with gradient norm penalty

Taxonomic Level	Classifier	Features	Validation Accuracy
Subfamily	SVM	Raw Pixels	95.0 (95.0)
	SVM	Multi-style deep features	**96.7 (96.7)**
	FC	Multi-style deep features	95.0 (95.0)
Tribe	SVM	Raw Pixels	89.3 (76.2)
	SVM	Multi-style deep features	**97.0 (96.3)**
	FC	Multi-style deep features	94.1 (89.4)
Genus	SVM	Raw Pixels	57.9 (51.8)
	SVM	Multi-style deep features	**80.3 (70.9)**
	FC	Multi-style deep features	79.8 (69.3)

Future Work

Connecting the generative and discriminative models. For example, using the generated images to train the discriminative model. Or to use the classifier in the discriminative model as part of the discriminator in the GAN. Partner with biologists to apply these models and methods to their classification paradigm.